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Effects of Near Infrared Light on Surgical Wound Healing: A Systematic Review and Meta-Analysis

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ABSTRACT

Near infrared (NIR) therapy is increasingly used to enhance postoperative wound healing, yet clinical trial results remain inconsistent. To evaluate the effectiveness of NIR therapy on postoperative wound healing and identify treatment parameters associated with optimal outcomes: This systematic review and meta-analysis registered at PROSPERO (CRD420251163415) assessed evidence on comparing NIR therapy (630–1100 nm) with standard care or placebo on healing of surgical-induced wounds. A multilevel random-effects meta-analysis of standardised mean differences (SMDs) was conducted. Moderator analyses examined the wavelength, fluence, session number, application technique and anatomical site. Risk of bias was assessed using Cochrane RoB 2.0 and certainty of evidence was rated with GRADE. Fifty-six trials ($N = 4920$) were included for systematic review and 35 trials contributed 69 outcomes to meta-analysis. NIR significantly improved wound healing (0.78, [0.46–1.09], $p < 0.01$) and reduced postoperative pain (0.71, [0.24–1.17], $p < 0.01$), but heterogeneity was high and effects varied across studies. Optimal outcomes were associated with short NIR wavelengths (700–850 nm), 4–10 sessions and non-contact application. Effects on swelling, scarring and inflammatory markers were inconsistent. Overall, certainty of evidence was very low. This first systematic review and meta-analysis indicates that NIR therapy demonstrates promise for enhancing postoperative healing and reducing pain, though effects vary by protocols.

1 | Introduction

Healing of surgery-induced wounds is a fundamental aspect of postoperative recovery. Effective healing reduces infection, scarring and recovery time, enabling patients to return to normal activities with reduced discomfort and improved quality of life. However, this process is often adversely affected by infection, ischemia, or excessive inflammation, which can delay tissue repair, resulting in hypertrophic or atrophic scarring, and even chronic non-healing wounds. Postoperative wound healing is a tightly regulated biological process progressing through overlapping stages of haemostasis, inflammation, proliferation and remodelling [1]. Haemostasis occurs immediately after tissue injury through vascular constriction and fibrin clot formation, which provides a provisional scaffold for cell migration and protection against infection [2]. During the inflammatory phase, neutrophils and macrophages remove debris and release cytokines that orchestrate subsequent tissue regeneration [3]. The following proliferative phase is characterised by fibroblast activation, collagen (type III) synthesis, angiogenesis and re-epithelialization to form granulation tissue [4]. In the final remodelling stage, extracellular matrix reorganisation, replacement of collagen III with collagen I, and myofibroblast-mediated contraction restore tensile strength and result in scar maturation [5]. Successful wound healing requires intact immune and inflammatory responses, sufficient perfusion and nutrition, and the absence of infection. Wound healing is frequently compromised in older or diabetic

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Key Points

- This systematic review and meta-analysis assesses the effects of near-infrared light (photobiomodulation) on surgery-induced wound healing, compared with sham or standard care.
- Overall, near-infrared light is associated with faster/improved postoperative wound healing and reduced postoperative pain, supporting its potential as a useful adjunct therapy after surgery.
- The observed benefits vary substantially across studies, suggesting that clinical outcomes strongly depend on treatment protocols, surgery and wound, and study design, and highlighting the need for better standardisation.
- Protocol-level trends suggest that more favourable outcomes may be linked to shorter NIR wavelengths (~700-850 nm), moderate treatment frequency (~4-10 sessions), and non-contact application, but the overall certainty of evidence remains limited and high-quality trials are still required to confirm optimal parameters.

patients and those with poor lifestyle habits, such as smoking or malnutrition [6, 7]. Impaired healing can prolong hospitalisation, elevate healthcare costs and induce chronic pain or disfiguring scars with significant physical and psychological burdens [8]. Conventional wound management strategies, such as dressings, antibiotic therapy and surgical revision, promote tissue repair but may have limited effectiveness in treating complex or infected wounds, particularly due to insufficient modulation of the wound microenvironment [9–11]. Importantly, many postoperative complications arise from disruption of specific healing phases—for example, prolonged inflammation, impaired perfusion/oxygenation and reduced fibroblast/keratinocyte activity—suggesting that adjunctive therapies that target these biological bottlenecks may enhance recovery. Consequently, there is an increasing interest in adjunctive modalities such as photobiomodulation (PBM) with near-infrared (NIR) light, which has shown potential to enhance postoperative wound healing by stimulating cellular energy metabolism, promoting fibroblast proliferation, modulating inflammation and thereby accelerating tissue recovery [1, 2, 12]. This mechanistic alignment with the major phases of repair (perfusion and inflammatory resolution early; proliferation and matrix remodelling later) provides a biological rationale for evaluating PBM specifically in postoperative wounds.

NIR, typically within the 630–1100 nm wavelength range, represents a non-ionising, non-thermal form of electromagnetic radiation. In PBM research, the clinically used 'therapeutic optical window' commonly spans red-to-near-infrared wavelengths, and in this review NIR generally refers to wavelengths at or above 700 nm while we report results across the full PBM window when trials include 630–699 nm [13–15]. NIR is a common PBM therapy to promote tissue repair, reduce inflammation and relieve pain [16]. Delivered through light-emitting diodes (LEDs) or low-power lasers, NIR light penetrates 2–3 cm into biological tissues without causing thermal damage, making it suitable for non-invasive clinical application [17]. The absorption of NIR photons by mitochondrial enzymes, particularly cytochrome c oxidase, enhances oxidative phosphorylation, increases ATP synthesis, induces nitric oxide (NO) release and triggers transient reactive oxygen species (ROS) signalling, collectively improving cellular metabolism and promoting tissue healing [18]. These primary photophysical events trigger downstream cellular responses, including fibroblast proliferation, collagen synthesis, angiogenesis and growth factor expression (e.g., TGF-β, PDGF and VEGF), which are vital for wound repair and tissue regeneration [19]. In postoperative wounds, NIR light accelerates the physiological healing cascade by stimulating fibroblast and keratinocyte proliferation, promoting collagen deposition and modulating acute inflammation, thereby facilitating faster wound closure and improved scar quality [20]. In contrast, chronic or ischemic wounds, such as diabetic ulcers, often exhibit mitochondrial dysfunction, oxidative stress and persistent inflammation [21, 22]. In chronic or ischemic wounds, such as diabetic ulcers, NIR therapy improves mitochondrial function, rebalances redox signalling and regulates inflammatory mediators—downregulating proinflammatory cytokines (e.g., TNF-α, IL-1β and COX-2) while upregulating anti-inflammatory ones (e.g., IL-10 and TGF-β)—and promotes macrophage polarisation from the M1 to M2 phenotype, thus reactivating impaired repair pathways [23]. The therapeutic outcomes of NIR largely depend on appropriate control of key parameters, including the wavelength, energy density and exposure duration, which must be optimised to maintain a stimulatory (rather than inhibitory) cellular response [24]. This dose–response behaviour (often described as biphasic) supports the need to identify clinically effective dosimetric ranges rather than assuming 'more light is better'. Overall, NIR light facilitates wound healing through mitochondrial reactivation, modulation of inflammatory and immune pathways, and enhancement of angiogenic and regenerative processes, establishing it as a safe, effective and clinically promising adjunctive therapy for both acute and chronic wound management [25].

Building on this physiological rationale, several clinical studies have applied NIR therapy in postoperative settings and have described improvements that mirror the mechanisms above—enhanced fibroblast activity, increased collagen deposition, faster epithelialization, reduced edema and better wound appearance [24, 26–28]. Recent postoperative studies in specific surgical contexts (e.g., caesarean incisions and oral/maxillofacial procedures) have similarly reported benefits in early wound scores, pain, or edema, although protocols and outcomes differ across trials [29–31]. To date, however, clinical outcomes across studies have not been uniform. Some trials report meaningful gains in wound closure time, early tissue regeneration, perfusion, or scar quality, whereas others find little or no effect on patient-important outcomes such as pain, infection, or overall healing rate [27, 32]. This apparent inconsistency does not contradict the biological data; rather, it likely reflects substantial heterogeneity across studies in (i) the NIR protocol used (wavelengths within or beyond the optimal window, different power outputs, total energy per session, continuous vs. pulsed delivery and number/timing of sessions), (ii) the populations enrolled (low-risk elective surgery

vs. patients with diabetes, obesity, or vascular compromise), (iii) the surgical procedures and wound types (orthopaedic, abdominal, breast, plastic/reconstructive, contaminated vs. clean) and (iv) the outcomes selected (time to complete epithelialization, dehiscence, surgical site infection, pain, edema, or cosmetic scales) [23, 25, 27, 33]. There are also indications from experimental work that excessively high doses or prolonged exposure may blunt or reverse PBM's beneficial effects, emphasizing the need to define clinically optimal dosimetric ranges [23, 25, 27, 33]. Because previous reports often combined acute postoperative wounds with chronic or ischemic wounds, or mixed red and NIR regimens, it is difficult to draw definitive conclusions for the specific context of postoperative healing. Therefore, a focused synthesis that isolates postoperative wounds and examines protocol parameters is needed to clarify effectiveness and guide clinical implementation.

Optimising postoperative wound healing remains a clinical priority, as delayed or impaired repair increases infection risk, prolongs hospitalisation and compromises patient quality of life. Establishing clear evidence on the therapeutic efficacy of NIR therapy could therefore have substantial implications for surgical care and rehabilitation. By synthesising clinical trial data across diverse surgical contexts, this study will help define the clinical potential of NIR therapy as a non-invasive, safe and cost-effective adjunct to conventional wound management. Furthermore, by analysing the influence of treatment parameters such as the wavelength, energy density, exposure duration and application frequency, this study aims to identify optimal dosimetric ranges and treatment conditions for maximising healing outcomes. Therefore, this study aims to conduct a systematic review and meta-analysis to quantitatively assess the effectiveness of NIR therapy on postoperative wound healing outcomes, including the wound healing rate, pain reduction, inflammation control and overall postoperative recovery as well as effective treatment parameters and dosage of NIR therapy on postoperative wound healing. To improve transparency and clinical interpretability, we additionally explore whether specific protocol features (e.g., the wavelength band, fluence, number of sessions and application techniques) are associated with larger effects. To the best of our knowledge, this is the first systematic review and meta-analysis on the effectiveness of NIR therapy on postoperative wound healing.

2 | Methods

2.1 | Protocol and Registration

This systematic review and meta-analysis was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [34]. The protocol was prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO) (CRD420251163415) and developed based on the Cochrane Handbook for Systematic Reviews of Interventions [35].

2.2 | Search Strategy

To guide the development of the search strategy, previously published reviews on photobiomodulation and wound healing were used to identify relevant keywords [14, 36–38]. Preliminary search strings were piloted and refined following the Peer Review of Electronic Search Strategies (PRESS) guideline to ensure a comprehensive and precise retrieval of potentially eligible studies [39]. A systematic literature search was conducted in the following electronic databases from their inception to October 2025: PubMed, Web of Science, Cochrane Central Register of Controlled Trials (CENTRAL), Scopus and CINAHL (via EBSCOhost). The search strategy combined Medical Subject Headings (MeSH) and free-text terms related to the concepts of 'near-infrared light' and 'postoperative wound healing'. In PubMed, the search string was structured as follows: ('infrared therapy' [MeSH] OR 'near infrared light' OR NIR OR 'low level laser therapy' OR LLLT OR photobiomodulation) AND ('wound healing' [MeSH] OR 'surgical wound' OR 'postoperative wound' OR incision OR 'tissue repair' OR 'scar formation'). Search terms were adapted for the syntax and thesaurus of each database. Boolean operators ('AND' and 'OR') were used to combine concepts, and the search was restricted to studies involving human participants, with no initial limits on publication year.

All identified records were imported into EndNote reference management software for the removal of duplicate citations. The titles of all retrieved records were then screened by one reviewer (J.L.) to exclude duplicates and clearly irrelevant studies. The abstracts of the remaining studies were independently reviewed by two authors (J.L. and B.E.). These initial stages were intentionally overinclusive. Subsequently, the full texts of the remaining articles were independently assessed for eligibility against the predefined inclusion and exclusion criteria by J.L. and V.G. Any disagreements at any stage of the screening process were resolved through discussion until a consensus was reached among the review authors. Finally, the reference lists of all included articles and relevant prior review papers were manually searched to identify any additional studies not captured by the electronic database search. All studies meeting the inclusion criteria in the systematic review were eligible for inclusion in the meta-analysis.

2.3 | Inclusion and Exclusion Criteria

Studies were selected according to predefined inclusion and exclusion criteria established prior to the search process. Studies were included if they: (1) were RCTs; (2) enrolled patients with surgical or postoperative wounds; (3) evaluated an intervention of NIR therapy, LLLT, or PBMT using LEDs or low-power lasers within the red to near-infrared spectrum (approximately 630–1100 nm); (4) included a control group receiving standard care, sham irradiation, or no light therapy; and (5) reported at least one relevant outcome (e.g., wound closure rate, pain, inflammation and scar quality). Only articles published in peer-reviewed journals in English were included, due to resource constraints for translation. Studies were excluded if they were: non-human studies; conference abstracts, reviews, or editorials; used light therapies outside the specified wavelength range; did not report postoperative wound outcomes; or lacked sufficient methodological detail or an accessible full text. All titles, abstracts and full texts were screened independently by two reviewers, and any disagreements were resolved through discussion or consultation with a third reviewer.

2.4 | Data Extraction

Data on the study population, intervention characteristics, comparison groups and outcome measures were independently extracted into a standardised form by two reviewers. Extracted study characteristics included the first author, year of publication, country, study design, sample size, participant demographics (age, sex, type of surgery and wound characteristics), and follow-up duration. Intervention details encompassed the light source type (laser or LED), wavelength, energy density (J/cm^2), power output, exposure duration, treatment frequency and total number of sessions. For control groups, data were collected on the type of comparator used, including standard wound care, sham irradiation, or placebo treatment. Where available, quantitative data on primary and secondary outcomes were extracted, including mean or percentage changes in wound healing rate, epithelialization time, pain score, inflammatory response, scar formation and overall postoperative recovery. For continuous outcomes, the mean change from baseline, the standard deviation (SD) of the mean change and the number of participants in each group at each assessment point were recorded. If studies reported multiple time points, only data from the longest follow-up period during which the NIR intervention was continued were extracted to ensure consistency. When the mean change from baseline was not directly reported, it was calculated from baseline and post-intervention values. Similarly, when the SD of change scores was not available, it was estimated using the formula: $SD_{change} = \sqrt{SD^2_{baseline} + SD^2_{post} - (2 \times Corr \times SD_{baseline} \times SD_{post})}$ where Corr was assumed to be 0.5, representing a moderate correlation between baseline and post-intervention measures. This approach, consistent with prior meta-analyses, was adopted to provide conservative estimates of variance [40, 41]. Sensitivity analyses were planned to assess the robustness of results using these imputed SDs [42, 43]. Unpublished data (mean and SD values) or clarification of ambiguous results were requested from corresponding authors when necessary. All extracted data were cross-verified by a second reviewer prior to inclusion in the quantitative synthesis.

2.5 | Data Coding

Descriptive data from each included study were coded for inclusion in the moderator analyses described later. In accordance with the objectives of this review, the parameters of the NIR or photobiomodulation intervention, study design and clinical outcomes were transformed into categorical or numerical variables. Two reviewers independently coded all data using a standardised form, and any discrepancies were resolved through discussion until consensus was reached.

2.5.1 | Intervention Moderators

Intervention parameters were prespecified and systematically coded as follows:

1. Wavelength: red (600–699 nm), short NIR (700–850 nm) and long NIR (851–1100 nm) per photobiomodulation classifications [17, 44].
2. Fluence (energy density, J/cm^2): low < 5, moderate 5–20, high > 20 in reference to published dosage guidelines [16, 45].
3. Power output: low < 100 mW, medium 100–500 mW, high > 500 mW following conventional classifications used in clinical PBM research [16, 46, 47].
4. Per-point irradiation time: short ≤ 30 s, medium 31–120 s, long > 120 s [37, 48].
5. Per-session irradiation time: short ≤ 300 s, medium 301–1200 s, long > 1200 s [47].
6. Application technique: contact (probe in direct touch with tissue) or non-contact (probe held ~0.5–10 mm from the wound).
7. Number of treatment sessions: short 1–3, medium 4–10, long > 10, reflecting typical clinical regimens.

These seven moderators were applied consistently in all subgroups, reflecting typical clinical regimens reported in the literature.

2.5.2 | Outcome Moderators

Outcome measures were grouped into domains representing the key clinical aspects of postoperative wound healing, informed by categorizations reported in previous reviews and further refined based on the outcomes identified in the included studies. Domains with sufficient study representation were retained for quantitative or descriptive synthesis and included: (1) wound healing and tissue repair, (2) pain and discomfort, (3) swelling, (4) sensory and neurological recovery, (5) immune response and (6) scar formation and cosmetic outcomes [32, 36, 38]. All included outcomes were continuous measures; no binary or time-to-event outcomes were identified. Accordingly, all pooled effects were expressed as standardised mean differences (SMD; Hedges g), computed from post-treatment means/standard deviations or change scores when reported. Scale directions were harmonised so that positive SMD indicates clinical improvement (e.g., faster healing, less pain and swelling, improved scar quality); outcomes where lower raw values reflect improvement (e.g., pain intensity, edema, inflammatory biomarkers, days to epithelialization, residual wound area) were inverted before pooling. When multiple assessment tools or time points were available, data from the longest follow-up within each domain were extracted. Domains comprised the following measures:

1. Landry Healing Index, REEDA scale, modified wound-healing index, epithelialization time (H_2O_2 test) and residual wound area (image analysis);
2. Patient-reported pain (VAS, NRS, VRS) and total analgesic consumption;
3. Edema quantified by standardised photographic or anthropometric facial measurements;
4. Light-touch testing, two-point discrimination and patient-reported sensory scores:
5. Laboratory biomarkers of inflammation/oxidative stress (e.g., cytokine concentrations);
6. Vancouver Scar Scale, Patient and Observer Scar Assessment Scale and quantitative colorimetry (ΔE analysis)

2.5.3 | Study Design Moderators

Moderator coding focused on key methodological and clinical factors that could influence the treatment effect. The control condition in each study was classified as 'standard wound care', 'placebo/sham irradiation'. Additionally, baseline wound characteristics were coded according to the anatomical location (e.g., facial, neck, trunk, limb, or donor site). These variables were systematically extracted to facilitate subgroup analyses assessing how differences in comparator condition and wound type might influence the outcomes. Because anatomical location in this dataset also distinguishes many mucosal wound contexts (e.g., oral/perineal) from cutaneous wounds (e.g., skin incisions and donor sites), this coding was used to account for clinically meaningful differences in tissue type and healing biology. We therefore interpret pooled effects as an average across heterogeneous wound contexts and emphasise location-specific findings where available.

2.6 | Risk of Bias Assessment

The methodological quality of all included randomised controlled trials (RCTs) was independently assessed by two reviewers using the Cochrane Risk of Bias 2.0 (RoB 2) tool [49]. This framework evaluates potential bias across five domains, including the randomization process, deviations from intended interventions, missing outcome data, measurement of outcomes and selection of reported results. Each domain was judged as having a low risk of bias, some concerns, or a high risk of bias, following the criteria outlined in the Cochrane Handbook for Systematic Reviews of Interventions [35]. The overall risk of bias for each study was determined by the highest level of concern identified among the assessed domains. All evaluations were conducted independently by two reviewers, and any disagreements were resolved through discussion or consultation with a third reviewer to achieve consensus. The results of the risk of bias assessment were summarised in both tabular and graphical formats to present the distribution of risk levels across the included studies.

2.7 | Statistical Analysis

Statistical analysis was conducted with R version 4.5.1 using the metafor package [50, 51]. For continuous outcomes, the standardised mean difference (SMD) was calculated using the mean change from the baseline, the standard deviation (SD) of the change, and the number of participants for both the NIR and control groups. If studies reported multiple outcome measures (e.g., wound healing rate, pain score and inflammation index) or multiple time points within the same participants, a three-level random-effects meta-analysis was fitted using a restricted maximum likelihood estimator to account for the dependency between these effect sizes [52]. Unlike traditional meta-analysis, which requires independence of effect sizes, this multilevel approach accounted for dependency by nesting effect sizes within studies, treating both the individual effects and the studies as random factors in the model [53]. The multilevel model was used to estimate an overall effect size of NIR interventions versus control on postoperative wound healing (Primary Objective). Heterogeneity was assessed using the Q statistic (with $p < 0.05$ indicating statistical significance) [54]. The I^2 statistic was also calculated to quantify the proportion of total variance due to heterogeneity, with values of 25%, 50% and 75% considered to represent low, moderate and high heterogeneity, respectively [55].

To investigate potential sources of variance, mixed-effects models were fitted to examine the moderators described in the Data Coding section [52]. Separate models were initially fitted to determine the main effects for each intervention parameter (e.g., wavelength, energy density), outcome domain (e.g., wound healing, pain) and study design moderator (e.g., control type and participant health status). The analysis of main effects was interpreted using the 95% confidence interval (CI) for the point estimates of each level of a moderator and the statistical significance of the omnibus test. Following the analysis of main effects, and in accordance with the study's objectives, significant moderators and those of key theoretical interest were added to a single model to examine possible confounding and to test for specific interaction effects (e.g., Wavelength × Outcome Domain) [52]. Funnel plots of the effect size against its standard error were visually inspected for small-study bias. Egger's test for funnel plot asymmetry was performed by modifying the multilevel random-effects model to include the standard error of the effect size as a moderator [56]. Because the primary synthesis used a three-level model (allowing multiple effect sizes per study), the funnel plot was constructed using the same set of extracted effect sizes entered into the multilevel model (i.e., one effect size per outcome domain per study at the longest available follow-up when multiple time points were reported). Accordingly, some studies contributed more than one data point to the funnel plot; statistical dependency was handled within the multilevel Egger regression. Small-study bias was considered present if the funnel plot showed visual asymmetry and the intercept from Egger's test was statistically significant ($p < 0.05$) [57].

The GRADE guidelines were applied independently by two review authors to evaluate the overall quality of evidence for the primary comparison of NIR therapy versus control [58]. The overall quality of evidence was initially rated as 'high' due to the inclusion of only randomised controlled trials. The evidence was downgraded by one level for each of the following: [1] overall risk of bias (if a substantial proportion of studies contributing to the outcome were rated as having a high risk of bias); [2] inconsistency (if $I^2 > 75\%$ and the 95% prediction intervals included the null effect); [3] imprecision (if the total number of participants was fewer than 400) [59]; and [4] publication bias (if indicated by a significant Egger's test). Indirectness was not considered a major concern due to the direct focus of the review's PICO question.

3 | Results

3.1 | Study Selection

The flow of records through the review is summarised in Figure 1. The initial search strategy returned 10 992 records, of which 5833 were screened at the title/abstract stage. One hundred four full-text articles were assessed for eligibility, and 48 full-text articles were excluded for the following reasons: 39 for

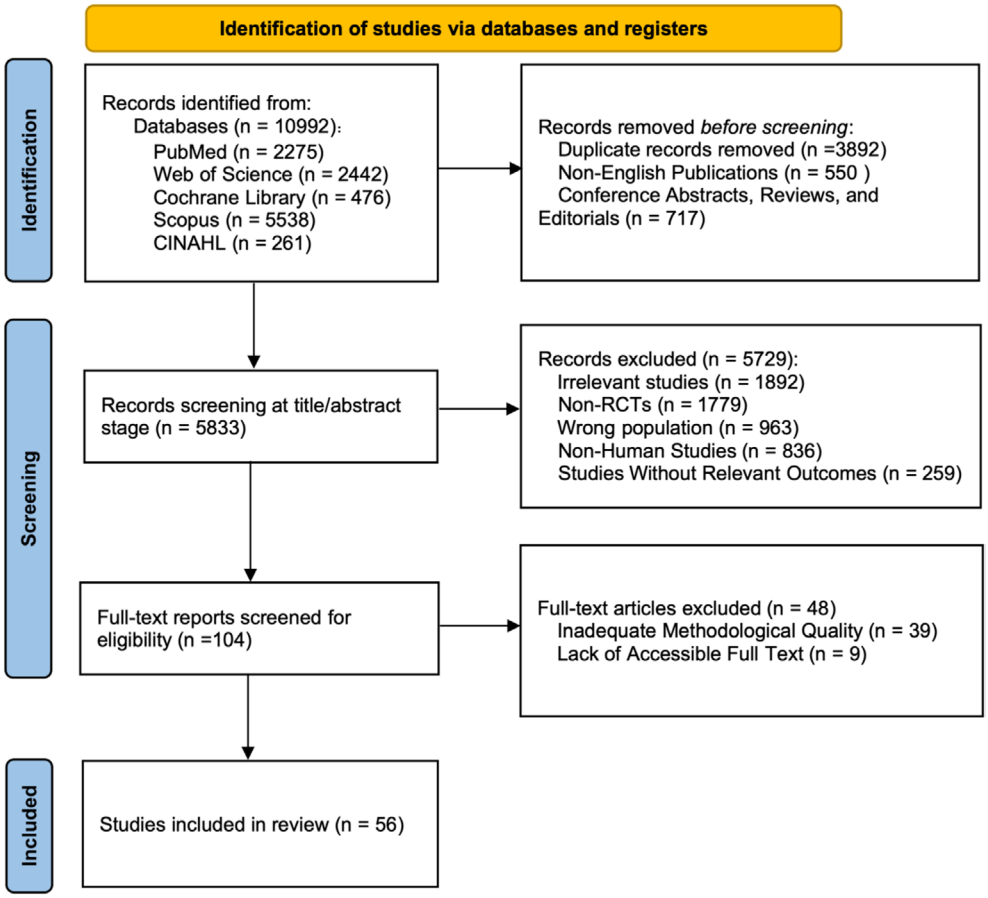


FIGURE 1 | PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) flow diagram.

inadequate methodological quality and 9 for lack of accessible full text. Finally, 56 studies met the inclusion criteria for the review. The methodological characteristics of these 56 included studies are summarised in Table S1 and the list of excluded 48 studies is presented in Table S2.

3.2 | Patient and Surgical Characteristics

Across the 56 RCTs included in this review, a total of 4920 patients were analysed. Most study populations consisted of generally healthy adolescents and adults undergoing routine postoperative recovery, with typical mean ages in the 20s–40s for oral and maxillofacial procedures (e.g., third molar extraction, periodontal surgery, free gingival graft harvesting, orthognathic surgery) [25]. Several trials enrolled older surgical populations, including patients undergoing coronary artery bypass grafting with median ages around 60 years [60–65], and individuals recovering from total hip arthroplasty or rotator cuff repair in middle to late adulthood [66, 67]. A small number of studies targeted special populations, such as postpartum patients after mediolateral episiotomy [9, 23, 68], and neonates following myelomeningocele repair [69]. Overall sex distribution varied by surgical model, with predominantly female cohorts in perineal and eyelid surgery trials [9, 23, 33, 68], predominantly male cohorts in cardiothoracic surgery [60, 61] and mixed-gender cohorts in most dentoalveolar and soft-tissue procedures [25, 27, 28, 66, 67, 70–102].

The surgical indications represented in the dataset were diverse and clinically relevant: oral and maxillofacial surgery (extraction sockets, periodontal graft donor sites, implant surgery, mandibular fracture repair, orthognathic surgery) accounted for the largest number of trials [25, 27, 28, 70–94]; followed by cardiothoracic procedures (sternotomy/chest and saphenous vein or thigh harvest sites in coronary artery bypass grafting) [60–65], abdominal and groin incisions (open gastroplasty, appendectomy, inguinal hernia repair) [24, 103, 104], perineal repair after childbirth episiotomy [9, 23, 68], orthopaedic and reconstructive procedures (rotator cuff repair, total hip arthroplasty) [66, 67], oculoplastic and facial aesthetic surgery (blepharoplasty, facelift incisions) [33, 105] and split-thickness skin graft donor sites [106]. Most studies, NIR therapy was initiated intraoperatively or within the first 24–48 h after surgery and then repeated over the first postoperative week (e.g., immediately, Days 1, 3 and 7), although some protocols extended treatment for longer courses—up to 2–4 weeks in sternotomy and thyroidectomy scars. Collectively, these trials therefore evaluate NIR therapy across a broad range of acute surgical wounds (incisional, graft donor sites) and across both low-risk and higher-risk healing environments.

3.3 | Endpoints and Outcome Assessment

Across the included randomised trials, postoperative outcomes were evaluated in several clinically relevant domains: wound

healing, pain, swelling/edema, scar/cosmetic appearance, inflammatory and tissue-regeneration markers, neurosensory recovery and functional recovery. Wound healing was assessed using validated clinical healing indices (e.g., the REEDA scale in perineal wounds and the Landry Healing Index in oral/periodontal wounds), incision-closure or epithelialization scores (including 0–3 closure grades, H_2O_2 bubbling tests of mucosal seal at graft donor sites) and quantitative morphometric methods such as percentage of remaining wound area measured from standardised photographs and computerised image analysis. Some studies also classified early wound complications (dehiscence, hematoma, seroma, local infection, delayed epithelial seal) as part of the healing endpoint.

Pain was typically measured using patient-reported visual analogue scales (VAS) or numeric rating scales (NRS), and in some trials supplemented by analgesic consumption or functional impact (e.g., chewing difficulty, reduced mouth opening, or limitations in activities of daily living). Swelling and edema were captured using linear facial distance measurements after oral and maxillofacial surgery, standardised edema scores, soft-tissue thickness measurements, or infrared thermography for local temperature and tissue inflammation.

Cosmetic and scar-related outcomes were evaluated using established scar and appearance scales, including the Vancouver Scar Scale, the Patient and Observer Scar Assessment Scale and structured assessments of scar pliability and tissue colour/ΔE, sometimes supported by 3D surface imaging. Inflammatory and regenerative responses were characterised biochemically, including local or systemic levels of cytokines (e.g., TNF-α, IL-6, IL-8), vascular and matrix remodelling markers (e.g., VEGF, MMP-8, MMP-9, TIMP-1) and, in cardiothoracic surgery cohorts, postoperative cardiac injury markers.

Neurosensory recovery after nerve-involving procedures (e.g., mandibular osteotomy, fracture repair) was assessed using standardised sensory tests such as Semmes–Weinstein monofilament light-touch detection, two-point and directional discrimination, thermal discrimination and patient-reported hypoesthesia or dysesthesia. Finally, several studies included site-specific functional outcomes: maximum interincisal opening and mandibular mobility after maxillofacial surgery; shoulder range of motion and disability scores (e.g., Constant–Murley, QuickDASH) after rotator cuff repair; sternum stability and ability to perform activities of daily living after sternotomy; and bone density or graft shrinkage from radiographic or photogrammetric analysis.

Most outcomes were recorded in the immediate and early postoperative period (within 24 h to Day 7), with some trials extending follow-up to 2–4 weeks or longer to assess scar quality, neurosensory recovery, structural stability (e.g., graft shrinkage, marginal bone loss), or return of function.

3.4 | Risk of Bias

Across 56 studies, overall judgements were 14/56 (25.0%) low risk, 27/56 (48.2%) some concerns and 15/56 (26.8%) high risk—indicating that most studies raised overall some concerns (Figure 2). By domain, missing outcome data and measurement of the outcome were predominantly low risk (82.1% and 78.6% low; 12.5% and 10.7% some concerns; 5.4% and 10.7% high, respectively), and selection of the reported result showed no high-risk ratings (28/56, 50% low; 28/56, 50% some concerns; 0% high). To improve transparency regarding Domain 4, we additionally summarised blinding practices: overall, 32/56 trials used a sham/placebo procedure (and therefore reported participant blinding), and 40/56 reported blinded outcome assessment. For pain outcomes ($n = 23$ trials), 14/23 reported participant blinding (sham), 15/23 reported blinded outcome assessment and 11/23 reported blinding of both participant and outcome assessor. In contrast, the randomization process and deviations from intended interventions showed more limitations (randomization: 24/56, 42.9% low; 27/56, 48.2% some concerns; 5/56, 8.9% high; deviations: 26/56, 46.4% low; 21/56, 37.5% some concerns; 9/56, 16.1% high). Concerns in the randomization process were mainly driven by insufficient reporting of sequence generation and, in particular, unclear allocation concealment. Limited blinding was more relevant to the measurement of outcomes, especially for subjective endpoints (e.g., pain ratings and clinician-assessed healing indices), and could also contribute to deviations from intended interventions when knowledge of group assignment plausibly influenced co-interventions, adherence, or protocol deviations. Taken together, the evidence base is best characterised as having overall 'some concerns' about bias, and pooled estimates should be interpreted with appropriate caution.

3.5 | Meta-Analysis of Postoperative Wound Healing Outcomes

In this multilevel random-effects meta-analysis, all outcomes were analysed as SMD (Hedges' g), with positive values favouring NIR. Thirty-five of the 56 included trials contributed data to the wound healing (69 effect sizes; Figure S1). The remaining 21 trials were not pooled because they lacked the statistics required to compute SMDs at the prespecified time point (e.g., missing SD/SE or otherwise incompatible reporting). The pooled effect favoured NIR with SMD = 0.78 (95% CI [0.46–1.09], $p < 0.01$), indicating a moderate-to-large improvement in postoperative healing. Heterogeneity was substantial (Cochran's Q, $p < 0.05$, $I^2 = 89.2\%$). Visual inspection of the funnel plot did not suggest marked asymmetry (Figure S2). However, funnel-plot assessment is limited and publication bias cannot be excluded. According to GRADE, the certainty of evidence supporting this effect was very low, downgraded for risk of bias, inconsistency and suspected publication bias (Table 1). This means that while the pooled results favour NIR on average, the true magnitude and clinical reproducibility across settings remain uncertain, and additional high-quality randomised trials are needed to confirm these findings.

3.6 | Moderator Analysis

To explore potential sources of variability, moderators were examined individually in separate models. The findings from this phase of analysis are summarised in Table 2 and discussed in the

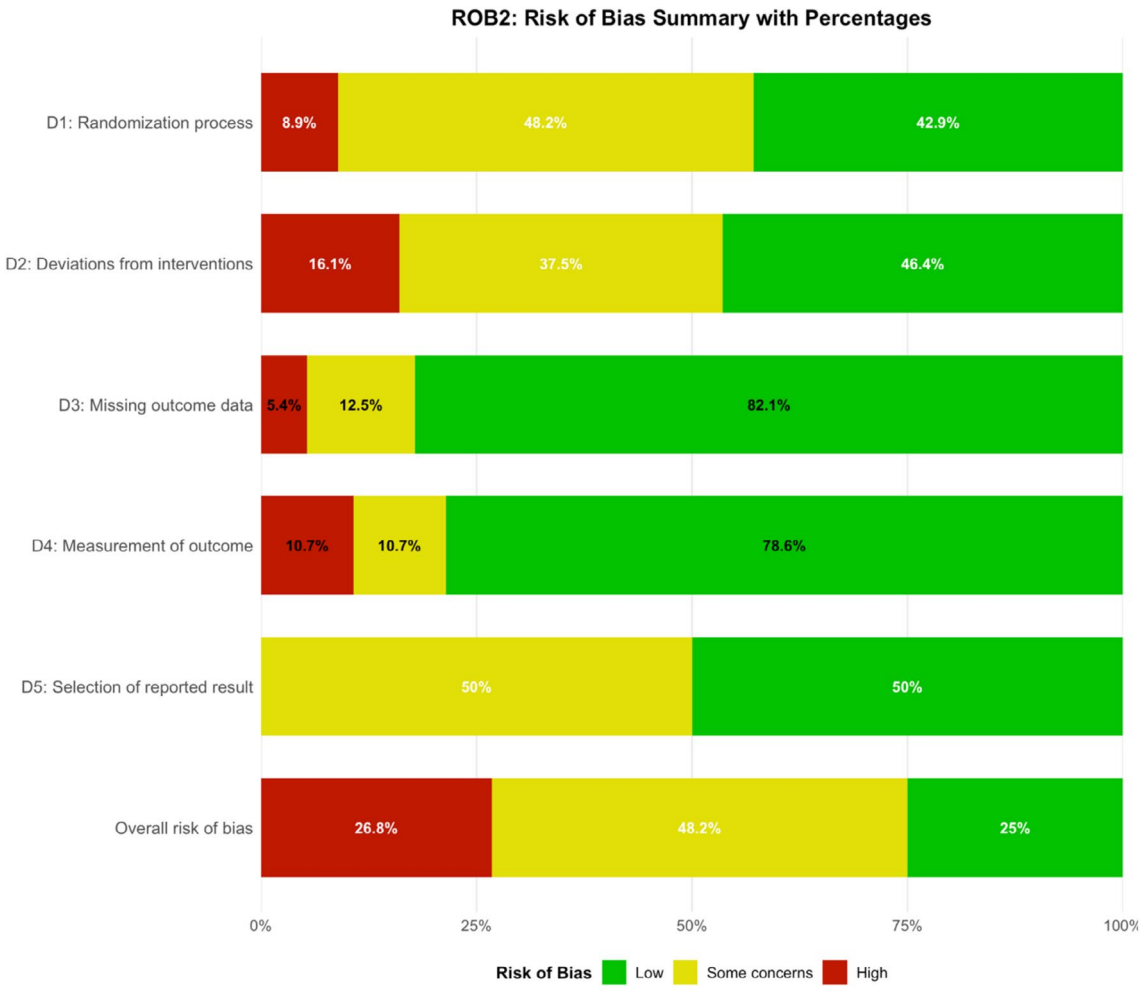


FIGURE 2 | Risk of bias assessment across included studies.

following section. Because these were univariable, exploratory moderator models, findings should be interpreted as hypothesis-generating rather than definitive evidence of effect modification.

3.6.1 | Intervention Moderators

Across intervention-related moderators, several factors influenced effect sizes; some parameters were associated with differences in pooled effect estimates.

1. Wavelength significantly moderated outcomes ($Q_4 = 14.88$, $p < 0.01$; Figure S3): short NIR showed the largest mean effect (SMD = 0.97, [0.36–1.57], $p < 0.01$), red light was also positive (0.72, [0.17–1.27], $p < 0.05$), long NIR was directionally positive but statistically non-significant (0.82, [−0.05–1.70], $p > 0.05$), and the combination of red light + short NIR was minimal (0.10, [−0.07–0.27], $p > 0.05$).
2. Fluence did not significantly moderate effects ($Q_3 = 2.05$, $p > 0.05$; Figure S4): low 0.50 ([0.12–0.88], $p < 0.05$), moderate 0.60 ([−0.04–1.23], $p > 0.05$) and high 0.74 ([−0.03–1.50], $p > 0.05$) overlapped broadly.
3. Power likewise was not a significant moderator ($Q_2 = 0.12$, $p > 0.05$; Figure S5), with low 0.42 ([−0.02–0.86], $p > 0.05$) and moderate 0.34 ([0.10–0.57], $p < 0.05$) showing similar magnitudes.
4. Per-point irradiation time did not significantly moderate effects ($Q_3 = 5.07$, $p > 0.05$; Figure S6), despite a numerically larger mean for long durations (short 0.54, [−0.04–1.11, $p > 0.05$]; medium 0.45, [−0.18–1.08, $p > 0.05$]; long 0.98, [−0.17–2.12], $p > 0.05$).
5. Per-session irradiation time also was not a significant moderator ($Q_3 = 2.91$, $p > 0.05$; Figure S7): short 0.57 ([0.22–0.93], $p < 0.05$), medium 1.92 ([−0.45–4.29], $p > 0.05$), long 1.09 ([−0.47–2.66], $p > 0.05$).
6. By contrast, number of sessions significantly moderated outcomes ($Q_3 = 11.01$, $p < 0.01$; Figure S8): medium programs produced the clearest benefit (1.13, [0.57–1.69], $p < 0.01$) relative to short (0.21, [−0.02–0.44], $p > 0.05$) and long (1.08, [−0.44–2.61], $p > 0.05$) schedules.
7. Application technique was also a significant moderator ($Q_2 = 10.78$, $p < 0.01$; Figure S9), with non-contact delivery (1.29, [0.71–1.87], $p < 0.01$) outperforming contact methods (0.30, [0.11–0.49], $p < 0.05$).

Overall, the wavelength, number of sessions and application techniques were statistically associated with differences in pooled

TABLE 1 | GRADE summary of findings.

Outcome	No. studies	Total participants	Effect (95% CI)	I^2 (%)	Egger *p*-value	Downgrade: Risk of bias (0/1/2)	Downgrade: Inconsistency (0/1/2)	Downgrade: Indirectness (0/1/2)	Downgrade: Imprecision (0/1/2)	Downgrade: Publication bias (0/1/2)	Final GRADE (high/mod/low/very low)
Pain and discomfort	23	1004	0.71 (0.24–1.17)	87.9	$p < 0.01$	1	2	0	1	0	Very low
Wound healing and tissue repair	23	792	0.70 (0.06–1.34)	88	$p < 0.01$	1	2	0	1	0	Very low
Swelling	11	513	0.85 (−0.09–1.78)	94.1	$p < 0.01$	1	2	0	1	0	Very low
Immune response	4	172	2.32 (−1.21–5.85)	94.3	$p < 0.01$	1	2	0	2	1	Very low
Scar formation and cosmetic outcomes	6	184	0.22 (−0.25–0.70)	30.4	$p < 0.01$	1	0	0	1	1	Very low
Sensory and neurological recovery	2	78	0.91 (−4.36–6.17)	65.5	$p < 0.01$	1	1	0	2	1	Very low

TABLE 2 | Results of moderator analysis.

Moderator	No. of effect sizes	Estimate mean (95% CI)	Prediction interval (95% PI)	*Q* statistic	Omnibus test of moderators
Intervention moderators					
Wavelength				$Q_{68} = 640.99$, $p < 0.01$	$Q_4 = 14.88$, $p < 0.01$
Red light	16	0.72 (0.17–1.27)	−1.13 to 2.57		
Short NIR	27	0.97 (0.36–1.57)	−1.92 to 3.86		
Long NIR	16	0.82 (−0.05–1.70)	−2.26 to 3.90		
Ret light + Short NIR	10	0.10 (−0.07–0.27)	−1.79 to 1.99		
Fluence				$Q_{52} = 548.46$, $p < 0.01$	$Q_3 = 2.05$, $p > 0.05$
Low	22	0.50 (0.12–0.88)	−0.90 to 1.90		
Moderate	16	0.60 (−0.04–1.23)	−2.00 to 3.48		
High	15	0.74 (−0.03–1.50)	−1.55 to 2.75		
Power				$Q_{45} = 139.34$, $p < 0.01$	$Q_2 = 0.12$, $p > 0.05$
Low	20	0.42 (−0.02–0.86)	−1.03 to 1.87		
Moderate	26	0.34 (0.10–0.57)	−0.59 to 1.27		
Per-point irradiation time				$Q_{34} = 194.55$, $p < 0.01$	$Q_3 = 5.07$, $p > 0.05$
Short	23	0.54 (−0.04–1.11)	−1.83 to 2.91		
Medium	10	0.45 (−0.18–1.08)	−1.26 to 2.16		
Long	2	0.98 (−0.17–2.12)	−1.14 to 3.10		
Per-session irradiation time				$Q_{61} = 628.83$, $p < 0.01$	$Q_3 = 2.91$, $p > 0.05$
Short	51	0.57 (0.22–0.93)	−1.83 to 2.91		
Medium	5	1.92 (−0.45–4.29)	−1.26 to 2.16		
Long	6	1.09 (−0.47–2.66)	−1.14 to 3.10		
Number of sessions				$Q_{68} = 640.99$, $p < 0.01$	$Q_3 = 11.01$, $p < 0.01$
Short	28	0.21 (−0.02–0.44)	−0.74 to 1.16		
Medium	35	1.13 (0.57–1.69)	−1.93 to 4.19		
Long	6	1.08 (−0.44–2.61)	−1.98 to 4.14		
Application technique				$Q_{60} = 572.21$, $p < 0.01$	$Q_2 = 10.78$, $p < 0.01$
Contact	26	0.30 (0.11–0.49)	−0.28 to 0.88		
Non-contact	35	1.29 (0.71–1.87)	−1.91 to 4.49		
Outcome moderators				$Q_{68} = 640.99$, $p < 0.01$	$Q_6 = 7.19$, $p > 0.05$
Pain and discomfort	23	0.71 (0.24–1.17)	−1.22 to 2.64		
Wound healing and tissue repair	23	0.70 (0.06–1.34)	−2.03 to 3.43		
Swelling	11	0.85 (−0.09–1.78)	−1.96 to 3.66		

(Continues)

TABLE 2 | (Continued)

Moderator	No. of effect sizes	Estimate mean (95% CI)	Prediction interval (95% PI)	*Q* statistic	Omnibus test of moderators
Immune response	4	2.32 (−1.21–5.85)	−3.17 to 7.81		
Scar formation and cosmetic outcomes	6	0.22 (−0.25–0.70)	−0.39 to 0.83		
Sensory and neurological recovery	2	0.91 (−4.36–6.17)	−4.44 to 6.26		
Study design moderators					
Control condition				$Q_{61} = 628.63$, $p < 0.01$	$Q_2 = 2.7$, $p > 0.05$
Placebo	30	0.60 (0.24–0.96)	−1.26 to 2.46		
Traditional care	32	1.11 (0.58–1.63)	−1.96 to 3.86		
Wound location				$Q_{64} = 629.40$, $p < 0.01$	$Q_4 = 16.51$, $p < 0.01$
Perineum and genital areas	7	0.13 (−0.13–0.40)	−1.01 to 1.03		
Oral cavity	49	0.71 (0.38–1.04)	−23.12 to 27.50		
Head and neck	8	1.29 (0.07–2.50)	0.26–0.53		
Donor sites for skin graft	1	1.33 (0.60–2.06)	NA		

effects in these univariable models, whereas the fluence, power and irradiation times showed no statistically significant moderation despite some numerical trends. Given substantial residual heterogeneity and very low certainty of evidence, these moderator findings should be viewed as preliminary signals that require confirmation in preregistered, adequately powered trials and in multivariable analyses where feasible.

3.6.2 | Outcome Moderators

The meta-analysis of outcome-related moderators revealed variability in treatment effects across clinical domains (Figure S10). The omnibus test of moderators indicated that overall variation among outcomes was not statistically significant ($Q_6 = 7.19$, $p > 0.05$), though the total heterogeneity remained significant ($Q_{68} = 640.99$, $p < 0.01$).

1. Pain and discomfort showed a moderate and consistent improvement (0.71, [0.24–1.17], $p < 0.01$), suggesting that interventions effectively reduced perceived pain. In Ye (2024) ($n = 73$ vs. 72), the pain score was -6.49 ± 2.57 in the NIR group versus -18.14 ± 5.05 in controls (difference +11.65 on the study scale) [33]. In Jaqueline de Oliveira Santos (2012) ($n = 26$ vs. 26), the perineal pain outcome was -1.50 ± 1.90 versus -2.50 ± 2.30 (difference +1.00) [23].
2. Wound healing and tissue repair demonstrated a comparable positive effect (0.70, [0.06–1.34], $p < 0.05$), but with substantial uncertainty and variability across study measures and contexts. In Antonio Scarano (2020) ($n = 10$ vs. 10), anti-desmin outcome was 12.75 ± 2.50 with NIR versus 4.00 ± 4.40 in controls (difference +8.75 on the study scale) [94]. In Revnak Metin (2018) ($n = 34$ vs. 34), defect volume density value was 333.09 ± 69.52 versus 297.20 ± 59.24 (difference +35.89) [84].
3. Swelling exhibited a small and highly variable effect (0.85, [−0.09–1.78], $p > 0.05$), implying inconsistent influence on edema reduction across studies. In Ye (2024) ($n = 73$ vs. 72), the mean swelling scores were -5.00 ± 1.42 in the NIR group versus -6.00 ± 1.33 in controls (difference +1.00 on the study scale) [33]. In Metin (2018) ($n = 34$ vs. 37), the edema values were -0.68 ± 0.64 versus -1.57 ± 0.77 (difference +0.89 on the study scale) [84].
4. Immune response showed the numerically largest mean effect (2.32, [−1.21–5.85], $p > 0.05$), but with wide uncertainty, reflecting sparse data and substantial heterogeneity. In Zou (2024) ($n = 35$ vs. 35), TNF-α outcome had a mean value of -19.23 ± 3.17 pg/mL in the NIR group versus -22.61 ± 5.82 pg/mL in controls (difference +3.38 on the study scale), while vascular endothelial growth factor (VEGF) in the same trial showed -24.26 ± 3.77 pg/mL versus -21.22 ± 0.68 pg/mL (difference −3.04). In El Makakey (2025) ($n = 20$ vs. 20), Metalloproteinases (TIMP-1) outcome was 2.08 ± 0.15 with NIR versus 1.52 ± 0.11 in controls (difference +0.56 on the study scale) [103].

5. Scar formation and cosmetic outcomes had a small, non-significant effect (0.22, [−0.25–0.70], $p > 0.05$), suggesting limited evidence of cosmetic improvement. In Kim (2022) ($n = 21$ vs. 22), the vascularity of thyroidectomy scar was 25.56 ± 2.39 with NIR versus 24.62 ± 3.17 in controls (difference +0.94) [107]. In Dias (2014) ($n = 16$ vs. 16), scar and tissue colorimetry outcome was -3.40 ± 1.10 versus -2.40 ± 1.10 (difference −1.00 on the study scale) [75].
6. Sensory and neurological recovery also displayed considerable uncertainty (0.91, [−4.36–6.17], $p > 0.05$) owing to few available comparisons and wide confidence intervals. In Sharifi (2020) ($n = 18$ vs. 18), pain discrimination outcome was 9.16 ± 0.92 in the NIR group versus 7.72 ± 1.16 in controls (difference +1.44 on the study scale) and two-point discrimination was 7.58 ± 1.19 versus 9.44 ± 1.93 (difference − 1.86) [80]. In Santos (2024) ($n = 22$ vs. 20), postoperative thermal outcome was 1.22 ± 2.35 with NIR versus 0.23 ± 1.18 in controls (difference +0.99 on the study scale) [108].

Overall, while between-category differences were not statistically significant, the results suggest that the pooled estimates are most favourable for pain and early healing indices; however, substantial heterogeneity and very low certainty of evidence mean these findings should be interpreted cautiously and may not generalise across all procedures, tissues, or protocols. Other outcomes showed weaker, non-significant, or highly imprecise effects.

3.6.3 | Study Design Moderators

Across study-design moderators, overall heterogeneity was high ($Q_{65} = 628.63$, $p < 0.01$). For the control condition (Figure S11), the omnibus test indicated no statistically significant differences among control types ($Q_2 = 2.7$, $p > 0.05$). Numerically, effects were larger in studies using traditional care as the comparator (1.11, [0.58–1.63], $p < 0.01$) than those using placebo (0.60, [0.24–0.96], $p < 0.01$). Wound location significantly moderated outcomes ($Q_{10} = 118.44$, $p < 0.01$; total heterogeneity $Q_{68} = 640.99$, $p < 0.01$; Figure S12).

Given established biological differences between cutaneous (skin) and mucosal wound healing, we present location results by broadly grouping sites that are predominantly cutaneous versus predominantly mucosal, while retaining the original anatomical categories used for coding [109]. Cutaneous/skin wound contexts showed several consistently favourable estimates. In the subgroup analysis, oral cavity wounds showed a clear benefit of NIR therapy (0.71, [0.38–1.04]; $p < 0.01$), whereas perineum/genital wounds showed little to no effect (0.13, [−0.13 to 0.40]; $p > 0.05$). Among predominantly cutaneous sites, effects were positive for head & neck wounds (1.29, [0.07–2.50]; $p < 0.05$) and for donor sites for skin graft (1.33, [0.60–2.06]; $p < 0.01$; based on a single study), while the trunk subgroup was imprecise and not statistically conclusive (1.45, [−0.26 to 3.17]). Overall, while control type did not significantly alter treatment effects, anatomical location did. In this dataset, several predominantly cutaneous sites (abdomen, inguinal region, chest and skin-graft donor sites) showed consistently favourable effects, whereas mucosal locations (perineum/genital, lip and oral cavity) tended to show smaller or highly imprecise estimates. These findings support cautious interpretation of pooled effects across tissue types and highlight the importance of reporting results by wound location when translating NIR therapy to specific clinical contexts.

3.7 | Sensitivity Analysis

Leave-one-out sensitivity testing demonstrated that the pooled effect of NIR therapy on wound healing was numerically stable (Figure S13). Sequential omission of each individual trial produced pooled SMDs ranging from 0.70 to 0.81, with all estimates remaining statistically significant ($p < 0.001$) and heterogeneity largely unchanged ($I^2 \approx 89\%$). Given the relatively similar statistical weights across individual trials, this limited fluctuation is expected and indicates that no single study exerted undue influence on the pooled estimate. No single study meaningfully altered the overall direction of effect, indicating that the observed benefit was not attributable to a single extreme outlier.

4 | Discussion

This systematic review and meta-analysis assessed the available clinical evidence on NIR therapy for improving postoperative wound healing across a wide range of surgical procedures. Although interest in NIR as an adjunctive modality has increased, individual trials have yielded variable findings and no consensus protocol has been established. By synthesising 56 RCTs and exploring intervention-, outcome- and study-level moderators, pooled estimates favoured NIR therapy for wound-healing/tissue-repair outcomes (0.70, [0.06–1.34], $p < 0.05$) and pain outcomes (0.71, [0.24–1.17], $p < 0.01$), but effects varied across studies and settings. Moderator analyses suggested larger pooled effects in studies using certain parameters (e.g., shorter NIR bands 0.97, [0.36–1.57], $p < 0.01$; medium treatment schedules 1.13, [0.57–1.69], $p < 0.01$; non-contact application 1.29, [0.71–1.87], $p < 0.01$), but these findings should be interpreted as exploratory and hypothesis-generating given substantial residual heterogeneity and very low certainty of evidence. However, the overall certainty of this evidence is low to very low because of risk of bias, substantial between-study heterogeneity and incomplete reporting. Accordingly, our findings should be interpreted as clarifying where and under what conditions pooled effects tended to be larger, rather than as definitive proof of efficacy in all postoperative settings.

4.1 | Clinical Outcomes of NIR Therapy

This meta-analysis is, to our knowledge, the most comprehensive synthesis of randomised controlled trials evaluating NIR therapy as an adjunct to postoperative wound care. Data from 35 trials (69 outcome estimates) showed that NIR therapy was associated with a moderate overall benefit compared with control conditions (standard postoperative care or sham/light-off comparators), with a pooled SMD of 0.74 ([0.42–1.06], $p < 0.01$). However, heterogeneity was substantial ($I^2 \approx 89\%$), indicating

that a large proportion of the variability in effect sizes reflects real between-study differences rather than chance; therefore, the pooled estimate should be interpreted as an average effect across diverse surgical contexts and NIR protocols, and its reliability for predicting benefit in any single clinical setting is limited. In the context of standardised mean differences, values in the range of 0.5–0.8 are usually interpreted as clinically relevant improvements, so an SMD of 0.74 suggests that, on average, NIR-treated wounds progressed through early healing faster than controls. This estimate should be interpreted cautiously because the overall risk of bias across included trials was most often rated as 'some concerns', and approximately one-quarter were rated high risk. The most frequent limitations involved the randomization process and deviations from intended interventions, commonly reflecting unclear allocation concealment and incomplete blinding, which can exaggerate pooled effects—particularly for subjective outcomes such as pain and clinician-rated healing indices. Given the high heterogeneity, the magnitude of this improvement is expected to vary across procedures, patient risk profiles and dosimetric regimens. This effect was most evident for outcomes that directly captured early wound repair, such as clinical healing indices (e.g., REEDA for perineal wounds, Landry Healing Index for oral/periodontal wounds), incision-closure or epithelialization scores and percentage reduction in wound area from serial photographs/ImageJ analysis. For these healing-focused endpoints, NIR favoured the intervention group with SMD = 0.70 ([0.06–1.34], $p < 0.05$), indicating earlier epithelial seal and better early tissue integrity than in control groups. Pain outcomes, typically measured with patient-reported VAS or NRS and sometimes supported by lower analgesic consumption or better mouth opening/functional use of the operated area, also improved with NIR (SMD = 0.71, [0.24–1.17], $p < 0.01$) relative to the same control conditions. Thus, the most consistently favourable pooled estimates were observed for wound-healing indices and pain relief, although variability across trials remained substantial. Other domains—swelling/edema (linear facial measurements, standardised edema scores), scar/cosmetic assessments (Vancouver Scar Scale, POSAS), and inflammatory or regenerative biomarkers—tended to show smaller or more variable effects, often with wide confidence intervals, suggesting that benefits beyond faster closure and analgesia may depend more strongly on wound location, the NIR wavelength band used, number of sessions, and timing of outcome assessment. This pattern is consistent with the observed heterogeneity, because these outcomes are more sensitive to procedural differences, baseline tissue injury and measurement timing than early closure and pain measures.

The present review confirms that the clinical evidence on NIR therapy for postoperative wounds is heterogeneous, but it also identifies clinical contexts and treatment parameters in which beneficial effects are more consistently observed. Clinical and translational studies have repeatedly shown that NIR can improve parameters of soft-tissue repair, inflammation and pain, but these benefits have not been uniform across all patient groups or surgical procedures [17, 18, 38]. The high I^2 suggests that heterogeneity is plausibly driven by both protocol-level factors and clinical differences across studies. Although subgroup and moderator analyses identified variables associated with larger pooled effects, heterogeneity remained high within most subgroups (often $I^2 > 75\%$). This indicates that the moderators examined explain only a portion of between-study variability and that clinical effectiveness remains difficult to predict across procedures, populations and protocols. Accordingly, pooled estimates should be interpreted as average effects across diverse contexts rather than as reliable estimates for any single clinical setting. In particular, small-to-moderate trials in oral and maxillofacial surgery (e.g., third molar extraction, periodontal/implant surgery, orthognathic procedures), cardiothoracic surgery (median sternotomy and saphenous vein/leg-harvest wounds) and minor dermatologic or plastic/oculoplastic incisions have reported shorter time to epithelialization or mucosal seal, better early graft/donor-site integration, less postoperative edema and reduced analgesic requirements in the first postoperative week when NIR was added to standard care [18, 38, 110]. These clinical signals are broadly consistent with perioperative and experimental observations of improved microcirculatory perfusion at wound margins, downregulation of proinflammatory cytokines such as TNF-α and IL-1β, and stimulation of fibroblast and keratinocyte proliferation [111]. Importantly, pooled estimates were most favourable for domains with the strongest mechanistic rationale—early structural repair and pain—while effects on swelling, scar quality and biomarker profiles were smaller, more variable, or imprecise, likely reflecting differences in PICOS elements across trials. In other words, heterogeneity likely reflects variation in treatment protocols (wavelength band, irradiance/fluence, session number/frequency, contact vs. non-contact delivery), patient characteristics (age, metabolic risk, smoking, baseline healing potential) and wound/surgical factors (anatomical site, tissue depth/vascularity, contamination risk and mechanical tension).

Importantly, several included trials reported minimal or no added benefit of NIR, and these neutral findings help explain the substantial between-study heterogeneity (Figure S1) [23, 25, 70, 72, 88, 93, 96, 97]. Across these trials, multiple study-specific factors were identified. First, some apparent improvements in pain were plausibly attributable to nonspecific effects of increased personal attention and wound-focused care (Hawthorne effect), rather than to the light exposure itself, resulting in similar symptom relief in both active and control arms [23]. Second, the irradiation parameters were likely subtherapeutic for the target tissue: for example, use of predominantly red light at approximately 660 nm, which penetrates only on the order of 0.5–2.5 mm, was described as unlikely to reach deeper muscle layers involved in episiotomy or other soft-tissue trauma [23]. Third, several trials explicitly acknowledged that the applied dosimetry was probably not optimal or not effective, limiting the biological impact of the intervention [9, 25]. Fourth, in bone-healing models, outcomes were sometimes assessed at delayed time points (e.g., 90 days), when both groups would be expected to show substantial spontaneous consolidation, potentially obscuring earlier differences; in addition, post-extraction sockets and other surgically created defects were described as inherently clinically favourable environments for new tissue formation, which may have reduced the measurable incremental benefit of NIR [72]. Fifth, some studies used outcome measures (e.g., radiographic density estimates) that were noted to be less sensitive than histomorphometric analysis of biopsy specimens, increasing the risk of a false-negative result [88, 93]. Finally, several neutral studies enrolled young, otherwise healthy patients with uncomplicated surgical sites and good baseline healing

potential (e.g., extraction sockets with ≥ 3 remaining bony walls), creating ceiling effects in which both groups healed well and between-group differences were intrinsically small [25, 70]. Together, these observations indicate that while the overall signal favours NIR therapy—particularly for accelerating wound closure and reducing early postoperative pain—the magnitude of benefit is influenced by patient factors, wound biology, dosing and wavelength parameters, choice and timing of outcome assessment, and the risk of nonspecific co-interventions. In addition, risk-of-bias limitations may have contributed to variability in observed effects. In trials with inadequate or unclear allocation concealment, baseline prognosis may differ between groups and incomplete blinding can increase performance and detection bias, especially when outcomes are patient-reported or depend on clinician judgement. By contrast, domains related to missing outcome data and outcome measurement were more often low risk, suggesting that attrition and measurement errors were less prominent threats than randomization and blinding. Accordingly, the high I^2 means that confidence should be placed more on the overall direction of effect than on the exact pooled magnitude, and clinical translation should emphasise protocol standardisation and procedure-specific evidence. This heterogeneity underscores the need for standardised, adequately powered protocols and for trials in higher-risk surgical wounds where meaningful delays in healing and uncontrolled pain remain clinically important.

Although our pooled analysis showed a consistent and clinically relevant benefit of NIR therapy for the two best-reported domains—postoperative wound healing and early pain reduction—the effects on other outcomes were clearly weaker and more uncertain when we examined the data across all included patient and surgical populations. The 35 trials that contributed 69 outcome estimates were drawn predominantly from generally healthy oral and maxillofacial surgery patients (third molar extraction, periodontal/implant procedures, orthognathic surgery, graft donor sites), with additional but smaller contributions from cardiothoracic (sternotomy and leg/vein-harvest wounds), abdominal/groin incisions, perineal episiotomy repairs, orthopaedic/reconstructive procedures, oculoplastic/facial aesthetic surgery and split-thickness skin-graft donor sites. Within this mix of acute, primarily closed surgical wounds, NIR produced statistically significant advantages for pain (0.71, [0.24–1.17], $p < 0.01$) and for healing/tissue-repair outcomes (0.70, [0.06–1.34], $p < 0.05$), but swelling/edema (0.85, [−0.09–1.78], $p > 0.05$), scar/cosmetic outcomes (0.22, [−0.25–0.70], $p > 0.05$), inflammatory/immune markers (2.32, [−1.21–5.85], $p > 0.05$) and sensory/neurosensory recovery (0.91, [−4.36–6.17], $p > 0.05$) were either small, highly variable, or based on sparse data. In other words, inside this specific postoperative evidence base—mostly low-to-moderate-risk surgical patients treated within 24–48 h after surgery—the most reliable clinical signal is for faster early repair and less pain, whereas benefits for edema, scar quality, biomarker modulation, or neurosensory outcomes cannot be considered established and appear to depend on wound location, NIR wavelength band, number of sessions and timing of assessment.

In the swelling subgroup, we pooled data from trials conducted mainly in oral and maxillofacial surgery (facial distance/linear measurements after dentoalveolar or orthognathic procedures) together with smaller numbers from abdominal/groin and cardiothoracic incisions. The pooled standardised mean difference favoured NIR (0.85, [−0.09–1.78], $p > 0.05$), indicating that, on average, studies reported less postoperative edema in NIR-treated sites than in control/sham groups. However, because the 95% CI crossed zero, this reduction did not reach conventional statistical significance, so it should be interpreted as a directionally beneficial but imprecise estimate. The wide interval almost certainly reflects (i) the modest number of studies contributing edema data, and (ii) marked heterogeneity in how swelling was measured (facial anthropometry, categorical edema scores, soft-tissue thickness, infrared thermography), which weakens comparability across surgical models. Biologically, an edema-reducing effect remains plausible: NIR/PBMT has been shown to temper acute inflammation, decrease microvascular permeability and improve lymphatic/tissue-fluid clearance, in parallel with downregulation of TNF-α, IL-1β and related inflammatory mediators [88]. A very similar pattern appeared in the inflammatory/immune response subgroup, where the pooled effect was numerically large but extremely imprecise (2.32, [−1.21–5.85], $p > 0.05$) because only four heterogeneous studies reported cytokine or matrix-remodelling markers at different postoperative time points. For sensory/neurosensory recovery, the evidence was even more limited (two studies only), and the pooled estimate was wide and non-significant (0.91, [−4.36–6.17], $p > 0.05$), so no reliable conclusion can be drawn. It is possible that the irradiation protocols in these trials did not match the anatomical course of the injured nerve or used wavelengths/doses less optimal for neural repair, but with such sparse and heterogeneous data these explanations should be regarded as hypotheses. Larger, procedure-specific RCTs with nerve-targeted NIR parameters and standardised neurosensory endpoints are needed to determine whether NIR can meaningfully improve postoperative sensory outcomes.

4.2 | Moderators of NIR Intervention Parameters

This meta-analysis systematically evaluated whether prespecified NIR therapy parameters moderated clinical effectiveness in postoperative wound healing. We examined seven intervention variables: (1) wavelength, (2) fluence, (3) power output, (4) per-point irradiation time, (5) per-session irradiation time, (6) number of sessions and (7) application technique. Of these, three parameters- (1) wavelength, (6) number of sessions, and (7) application technique—were statistically significant moderators of treatment efficacy. The remaining parameters- (2) fluence, (3) power output, (4) per-point irradiation time and (5) per-session irradiation time—did not significantly moderate outcomes within the ranges used in the included trials. Taken together, these moderator findings may help explain why some trials reported larger benefits while others reported little or no added effect; however, given very low certainty of evidence and high heterogeneity, these associations should be regarded as provisional.

4.2.1 | (1) Wavelength

Wavelength was a significant moderator of treatment efficacy. Interventions delivered in the short near-infrared range

(≈700–850 nm) were associated with the largest improvements in clinical outcomes. This is consistent with several mechanistic and clinical observations from the included studies. First, penetration depth differs meaningfully across wavelengths: red light around 660 nm typically reaches only ≈0.5–2.5 mm into soft tissue, whereas infrared light in the ≈780–810 nm range can penetrate on the order of 8–10 mm into deeper layers such as muscle [23]. Some trials that reported minimal benefit explicitly attributed this to inadequate penetration of the wavelength used; for example, one pilot study noted that the red light applied 'was probably not powered to reach the muscle layers affected by the episiotomy' and proposed shifting to an infrared wavelength (around 780 nm) in future work to reach the relevant damaged tissues [23].

Second, several protocols intentionally selected wavelengths in the 800–850 nm band to increase absorption by cytochrome c oxidase, thereby enhancing mitochondrial activity and downstream repair processes, compared with earlier protocols that used lower wavelengths (e.g., 637–660 nm) or older 780 nm settings [25]. This reflects an evolution from early low-level laser approaches (e.g., He-Ne and lower-power diodes) toward higher-wavelength gallium–aluminium–arsenide systems (≈800–810 nm) that deliver energy more effectively to deeper surgical sites. Notably, multiple studies also compared single infrared wavelengths (e.g., 808 or 810 nm) to dual-wavelength protocols combining infrared and red light (e.g., 808/660 nm or 810/660 nm). The rationale for dual-wavelength PBMT is that superficial red light supports epithelial and flap closure, whereas infrared reaches deeper tissues and modulates the inflammatory response; some authors suggested this could be synergistic for simultaneous superficial and deep healing [70, 103]. However, several head-to-head comparisons reported that adding the superficial red component did not produce a statistically significant additional benefit over infrared alone in certain surgical contexts (e.g., third molar extraction), and hypothesised that this is because the infrared component already exerted the dominant anti-inflammatory and tissue-stabilising effect at depth [72, 73]. That said, the same studies noted that red wavelengths may still be advantageous in situations where superficial closure is compromised (e.g., flap tension, mucosal edge integrity) and cautioned that wavelength choice may need adjustment in patients with darker skin to ensure effective delivery.

Taken together, these findings help explain heterogeneity in the overall literature: trials using shorter-penetrating red light alone in anatomically deeper injuries were more likely to report little or no added clinical benefit, whereas trials using wavelengths in the ≈780–850 nm infrared range—capable of deeper penetration and stronger mitochondrial absorption—were more likely to report meaningful improvements in healing and symptom control.

4.2.2 | (2) Fluence (Energy Density, J/cm^2)

Fluence did not emerge as a statistically significant moderator of clinical outcomes in our meta-analysis, but this should be interpreted with caution rather than as evidence that fluence is unimportant. Across the included trials, energy densities varied from very low single-application exposures (e.g., ≈2–4 J/cm^2 or ≈6 J/cm^2 delivered by low-power diode lasers) to substantially higher regimens (30–60 J/cm^2 and above). Several studies reported little or no measurable benefit at lower doses. For instance, López-Ramírez et al. found no beneficial effect of a single LLLT exposure at 810 nm, 0.4 W for 32 s, with an energy density of 4 J/cm^2 [86]; similarly, classic work using a soft laser with ≈2.2 J/cm^2 energy delivery (50 mW) did not detect significant improvement in full-thickness skin defects [28]. Other periodontal and oral surgery studies have likewise reported that diode laser application at ≈4 J/cm^2 did not enhance epithelial regeneration, vascular formation, or early wound quality [75, 81]. By contrast, several studies have documented clinically meaningful benefits at higher fluences. In palatal donor sites, NIR at 15 J/cm^2 accelerated wound closure after connective tissue graft harvesting compared with no laser [9]. In a direct comparison of two regimens (30 J/cm^2 vs. 60 J/cm^2), the 60 J/cm^2 protocol resulted in a statistically smaller residual wound area at postoperative Day 7 than both 30 J/cm^2 and sham, indicating a possible dose–response in early mucosal repair [73]. Preclinical work in cutaneous wounds also supports this pattern: doses of 50 J/cm^2 and even 72 J/cm^2 have been associated with reduced inflammatory infiltrate, increased collagen organisation and accelerated matrix maturation within the first postoperative week [104]. One proposed explanation is that higher incident fluence may be necessary to compensate for attenuation of light by reflection, absorption and scattering in overlying tissue, so that an adequate amount of energy actually reaches the biologically relevant target layer [84]. At the same time, several clinical protocols in soft tissue grafting and FGG surgery have reported favourable or at least biologically plausible effects with intermediate dosing (e.g., 6 J/cm^2 NIR adjunct to grafting, 10 J/cm^2 at 50 mW every other day for 15 days, or ≈7.5 J/cm^2 soft-tissue diode laser), and some studies have suggested improved graft stability, reduced shrinkage at 6 months, or enhanced early revascularization at doses in the ≈3–10 J/cm^2 range [85, 86, 112].

Taken together, these findings illustrate two points that help interpret our moderator analysis. First, most human RCTs clustered within what is already considered a 'therapeutically reasonable' dosing window (often ≈3–10 J/cm^2 per application, occasionally higher), which limits between-study contrast and makes it statistically difficult to detect fluence as a moderator. Second, both very low single-shot exposures (≤4 J/cm^2) and inadequately powered applications have repeatedly produced null or minimal effects, whereas higher delivered fluence (15–60 J/cm^2 and above), or repeated applications of mid-range fluence, has been linked to faster early closure, reduced inflammation and improved matrix organisation in mucosal and soft tissue surgical wounds. Thus, the absence of a significant moderator effect for fluence in our pooled analysis likely reflects a combination of restricted dosing ranges across trials and heterogeneous reporting, rather than evidence that energy density is clinically irrelevant.

4.2.3 | (3) Power Output

Power output did not emerge as a statistically significant moderator of clinical outcomes in our meta-analysis ($\chi^2 = 0.12$, $p > 0.05$). However, as with fluence, this finding should not be

interpreted to mean that output power is clinically irrelevant. Across the included trials, lasers were delivered over a fairly constrained therapeutic range, generally in the low-power, non-thermal NIR range and many studies reported favourable healing, edema control, or pain reduction using devices between roughly 50 and 200 mW. For example, several RCTs applied NIR at 808–810 nm with output powers of 100 mW (delivering 6–12 J over three intraoral points) or 200 mW (6 J per point), and both protocols reported clinically satisfactory postoperative outcomes [25]. A review similarly suggested that wavelengths between 780 and 810 nm combined with power levels up to approximately 200 mW were associated with improved pain and edema control, and the authors of one trial specifically noted that they achieved 'very satisfactory results' using a 100 mW device even though higher-power devices had been proposed as optimal [36]. Other studies in oral and maxillofacial surgery, graft donor site healing, or soft tissue repair used diode lasers at 40–50 mW, 50 mW or 100 mW, and in some cases increased power from 20 to 50 mW (together with longer irradiation time) to raise delivered energy per point to ≈4 J, yielding a total energy delivery nearly 20-fold higher than in the earlier version of the protocol [72, 93, 100]. There are also reports using higher absolute output in other surgical contexts, including 0.3–0.4 W continuous diode delivery and even 1.5–2.5 W in Nd:YAG-based protocols, although these higher-power approaches are typically coupled with short application times, targeted spots and explicit attention to avoiding thermal injury [72, 86, 90].

Two points follow from these observations. First, within the clinically common 'low-power PBM' band (tens to a few hundred milliwatts, or low-watt lasers applied in short bursts), many protocols can deliver biologically active total energy, either by increasing power, increasing exposure time, or both. As a result, different studies can achieve comparable total energy deposition at the tissue despite using different nominal output powers. This interchangeability makes it statistically difficult to detect power output itself as an independent moderator. Second, trials that reported little or no additional benefit frequently used low-power single exposures that delivered only a few joules in total, whereas protocols that escalated either the output (e.g., 100–200 mW per point) or the cumulative delivered energy per session tended to report improvements in pain, edema, or early soft tissue closure. In that sense, heterogeneity in reported clinical benefit across studies may reflect differences in how effectively power output was translated into delivered energy at the target tissue, rather than the absolute wattage or milliwatt rating of the device alone.

4.2.4 | (4) Per-Point Irradiation Time and (5) Per-Session Irradiation Time

Neither per-point irradiation time nor per-session irradiation time significantly moderated clinical outcomes in our meta-analysis. This does not imply that timing is unimportant; rather, it reflects the fact that irradiation time is tightly coupled to other parameters (power, fluence, total energy per point), and that published protocols use a wide range of exposure durations without a clear consensus.

At the per-point level, several studies explicitly report adjusting irradiation time in order to deliver a higher energy dose per point [64, 94, 106]. For example, one protocol increased the exposure time from 10 to 80 s and increased the output power from 20 to 50 mW, which resulted in an energy delivery of 4.0 J at each point [94]. These studies illustrate two important features of clinical PBM practice: (i) 'dose' at a point on the wound is defined jointly by power and exposure time, and (ii) increases in irradiation time are often used to compensate for lower power, or vice versa, to reach a target energy per point [38]. Because power and time are intentionally co-varied to achieve a desired per-point energy, it becomes statistically difficult to isolate irradiation time itself as an independent driver of efficacy in moderator analysis.

At the per-session level, reported exposure durations span orders of magnitude across indications, and are often delivered repeatedly over days to weeks. Examples from the broader NIR literature include sessions lasting 2.5 min in continuous mode, 10 min per day for 5 consecutive days, 15 min twice weekly for 3 weeks, or 20 min a day over 5 days [113]. A review of irradiation parameters across studies reported that stated irradiation times ranged from as short as 1 s to as long as 7000 s (approximately 1 h 56 min), with an average of about 766 s (≈13 min), and noted that very long continuous exposure times can become impractical [110]. That same review emphasised that irradiation time is one of the components used to calculate total dose, and that 'the higher the power output the lower the exposure time', underscoring again that duration is rarely manipulated in isolation [114].

In this context, the absence of a detectable moderating effect for per-point irradiation time or per-session irradiation time in our meta-analysis likely reflects two constraints: first, clinically used exposure times already fall within ranges considered acceptable and biologically active; and second, changes in irradiation time are typically paired with changes in power or total delivered energy.

4.2.5 | (6) Number of Sessions

The number of treatment sessions significantly moderated clinical outcomes in our meta-analysis ($\chi^2 = 11.01$, $p < 0.01$; 0.74, [0.42–1.06], $p < 0.01$). Protocols delivering a moderate total number of sessions (approximately 4–10 applications over the postoperative period) were associated with larger effect sizes than protocols using only a single exposure or very prolonged/high-frequency regimens (1.13, [0.57–1.69], $p < 0.01$). This pattern is consistent with the general biphasic (hormetic) response model in NIR, in which too little stimulation fails to engage a meaningful biological response, while excessive or prolonged stimulation may lead to diminishing returns [111].

Evidence from the broader NIR literature illustrates how heterogeneous dosing schedules are in practice. Some studies used a single application only—for example, cells or wounds exposed once to laser light (e.g., ~97 mW/cm^2 for 2 min immediately following injury) and then followed for up to 48 h without retreatment [38]. Other studies delivered repeated treatments at tightly defined intervals: in an acute retinal injury model, animals received three brief NIR-LED exposures at 5, 25 and 50 h; in toxic or inflammatory models, light was delivered at regular 24-h intervals; and in paediatric bone

marrow transplant settings, NIR was administered once daily for 14 consecutive days post-transplant [113]. Additional wound-healing and soft-tissue protocols in both animal and human studies have used once-daily treatment, every-other-day treatment, several times per week, or 'daily until moderate improvement was observed', reflecting substantial variability in clinical practice [110]. Importantly, despite this variability, most regimens still fall into one of two extremes: either a single-session exposure (under-treating, biologically speaking) or intensive multi-day/multi-week courses (potentially risking overtreatment). Our moderator finding helps interpret why individual RCTs sometimes disagreed on clinical benefit. Studies relying on a single exposure or very sparse dosing were more likely to report little or no added clinical effect, especially for outcomes like pain reduction or edema control [67, 70, 86, 98, 102]. By contrast, protocols that delivered repeated but time-limited courses—on the order of several applications across the early postoperative window—more often reported measurable improvements in wound closure, inflammation control and symptom relief [23, 24].

Taken together, these observations suggest two conclusions. First, 'number of sessions' is not just a logistical parameter; it is biologically active. Repeated exposure over the first several days of healing appears to be important for driving a sustained response, particularly in modulating inflammation and supporting early tissue repair, whereas a single application is often insufficient. Second, there is no universally accepted session frequency across indications—studies have used schedules ranging from three discrete treatments within 50 h to once daily for 14 days—so our finding that a moderate number of sessions (4–10 total applications) was associated with greater clinical benefit provides quantitative support for a practical middle ground. This helps explain part of the heterogeneity in overall effect size in our meta-analysis: trials that dosed NIR repeatedly but not indefinitely tended to contribute to the positive pooled signal, whereas trials with only one exposure or highly prolonged/intensive regimens contributed more variable or attenuated effects.

4.2.6 | (7) Application Technique

Application technique was a significant moderator of clinical outcomes in our meta-analysis ($\chi^2 = 10.78$, $p < 0.01$). Specifically, non-contact delivery techniques were associated with greater clinical benefit than contact-based techniques (1.29, [0.71–1.87], $p < 0.01$). In non-contact protocols, the light source is positioned at a controlled distance from the tissue rather than pressed directly against it (0.30, [0.11–0.49], $p < 0.05$). For example, in one study a 3 mm fibre was coupled to the laser output and held approximately 10 mm above the surface of a 35 mm culture dish, directing the beam toward the centre without touching the target [75]. Similar approaches in wound models report the probe or LED held several millimetres above the wound bed, allowing irradiation of the site without mechanical contact. By contrast, other studies explicitly used contact mode, describing the tip of the laser probe as being held gently in direct contact with the wound (often through a transparent dressing) to deliver energy directly to the tissue surface. These technique differences are not purely procedural; they have biological and clinical implications that help explain heterogeneity in the literature. Non-contact application minimises direct mechanical disturbance of fresh surgical tissue, avoids pressure on inflamed or sutured wound edges and reduces the risk of local irritation or contamination [114]. Maintaining a small standoff distance also facilitates more uniform photon distribution across irregular wound geometries, rather than concentrating energy at a single point of probe contact [114]. In contrast, contact-mode delivery may introduce slight compression, disrupt fragile margins (e.g., graft donor sites, mucosal closures), or introduce variability depending on how firmly the operator 'touches' the wound [32].

Notably, prior narrative reviews have described both approaches—contact and non-contact—and have documented a wide range of implementations ('laser probe 10 mm above wound', 'tip in contact with wound through dressing', 'distance of LED from wound not reported') without reaching a conclusion about which is superior [115]. Our moderator analysis adds quantitative evidence to that discussion: trials using non-contact delivery tended to contribute larger effect sizes for clinically relevant outcomes such as early wound closure, edema control and pain reduction, whereas trials relying on direct-contact application were more likely to report attenuated or null effects. In this context, application technique appears to act as more than a methodological detail; it likely influences how consistently therapeutic light actually reaches the target tissue without introducing secondary mechanical or infectious stressors. This supports the practical interpretation that, in acute postoperative wounds and inflamed surgical fields, controlled non-contact delivery should be considered a preferred approach for clinical NIR.

These moderator analyses clarify why the overall pooled effect of NIR therapy on postoperative healing was both clinically meaningful (0.74, [0.42–1.06], $p < 0.01$) and highly heterogeneous. Trials that used (i) short-wavelength NIR in the 700–850 nm range, (ii) a moderate number of treatment sessions [4–10] and (iii) non-contact application tended to yield the largest improvements in wound healing rate and early pain reduction. By contrast, trials that deviated from these conditions—very low or very high dosing frequency, suboptimal wavelength selection, or contact-based delivery—were more likely to produce small or null effects. In this way, differences in parameter choice can account for much of the apparent disagreement in the literature, where some studies report dramatic improvements in healing and symptom control while others report minimal or no added benefit. Overall, our results suggest that not all 'negative' trials should be interpreted as evidence against NIR therapy itself. Rather, variation in wavelength, dosing schedule and delivery technique appears to meaningfully shape clinical outcomes. These moderator analyses help explain why the overall pooled effect of NIR therapy on postoperative healing was directionally favourable (0.74, [0.42–1.06], $p < 0.01$) yet highly heterogeneous. Trials that used (i) short-wavelength NIR in the 700–850 nm range, (ii) a moderate number of treatment sessions [4–10] and (iii) non-contact application tended to report larger pooled effects, whereas trials that deviated from these conditions (e.g., single exposure, suboptimal wavelength selection for deeper targets, or contact-based delivery) more often reported smaller or null

effects. However, these patterns should not be interpreted as definitive protocol recommendations for routine clinical use, because residual heterogeneity remained high and the certainty of evidence was very low. Instead, they provide testable hypotheses for future preregistered RCTs—namely, that wavelength band, session number and application technique may be important design features to standardise and evaluate in procedure-specific studies.

4.3 | Moderators of Study Design Characteristics

The type of control condition did not significantly moderate the observed effects ($Q_3 = 2.7$, $p > 0.05$). While numerically larger effects were found in comparisons with traditional care (0.95, [0.41–1.48], $p < 0.01$) than with placebo (0.60, [0.19–1.01], $p < 0.01$), the overlap of confidence intervals and non-significant omnibus test suggest these differences should be interpreted with caution. In one such study of acute postsurgical wounds, NIR was compared with microcurrent therapy, which itself is intended to modulate healing biochemistry [103]. In that comparison, the pooled effect size was smaller and imprecise (0.28, [−0.35–0.90], $p > 0.05$), limiting any firm conclusion about the superiority of NIR over another bio-stimulatory intervention in this specific patient population. Because only one RCT contributed data for this comparator, the estimate is underpowered and should be viewed as exploratory. These findings suggest that pooled effects tended to favour NIR across different comparator conditions, but differences by control type were not statistically supported and should be interpreted cautiously. However, the lack of significant moderation by control type also reinforces the methodological variability in trial designs, and underlines the importance of transparent, standardised reporting of control conditions in future studies.

By contrast, anatomical location significantly moderated treatment effects ($Q_{10} = 118.44$, $p < 0.01$), with several sites demonstrating consistent and clinically relevant improvements. Notably, donor sites for skin grafts (1.33, [0.60–2.06], $p < 0.01$), the chest (1.06, [0.29–1.83], $p < 0.01$), abdomen (0.66, [0.29–1.04], $p < 0.01$) and inguinal region (0.39, [0.25–0.52], $p < 0.01$) yielded the most robust effects. These locations may represent more favorable optical and biological conditions for NIR penetration and response, including adequate vascularization, relatively superficial tissue depth and limited structural obstruction [14]. At the same time, it is important to interpret these location effects in light of well-described tissue-level differences between cutaneous and mucosal healing. Oral mucosa is generally thicker and has a higher baseline epithelial proliferation capacity than skin, which contributes to faster closure and less scarring in many models. Oral wounds also heal in a continuously moist, saliva-exposed environment and saliva contains multiple bioactive factors that can stimulate repair [109]. Consistent with these intrinsic differences, oral wounds have been observed to close faster than skin wounds across species, while skin wounds can show more persistent angiogenesis that is linked to hypertrophic scarring [109]. These baseline contrasts offer a biologically plausible explanation for why some mucosal or mucocutaneous subgroups in our dataset showed smaller or less consistent incremental benefits, including minimal effects in the perineum and genital areas (0.01, [−0.60–0.61]) and lip (0.08, [−0.11–0.27]) and why the oral-cavity location subgroup showed wide uncertainty, likely reflecting heterogeneity in procedures, outcomes and follow-up timing within a tissue type that already tends to heal rapidly. Several anatomical subgroups, such as the eyelid (3.84, [−8.32–16.01], $p > 0.05$) and oral cavity (2.19, [−22.55–26.94], $p > 0.05$), exhibited wide confidence intervals and high uncertainty, likely due to sparse data or inconsistent parameter reporting. These imprecise estimates preclude firm conclusions but highlight areas where further research is needed. Previous reviews have similarly noted that anatomical and tissue-specific factors, including chromophore density, optical properties and inflammatory state, may influence treatment response, supporting the relevance of anatomical site as a key moderator [116].

The present analysis indicates that anatomical location is an important moderator of NIR therapy efficacy in postoperative wound healing, and that the control condition did not significantly alter the observed effect sizes. These findings suggest that NIR benefits are not restricted to a specific comparator and that tissue-level factors such as depth, perfusion and local inflammatory environment contribute meaningfully to treatment response. Given the established biological differences between skin and mucosal repair, we emphasise that 'anatomical location' in this review also functions as a pragmatic proxy for tissue context, cutaneous versus mucosal or mucocutaneous, and results should be generalised primarily within comparable tissue environments rather than assumed to transfer uniformly across them. At the same time, these results must be interpreted in the context of study quality: several included trials were limited by small sample sizes, incomplete blinding, unclear allocation methods and heterogeneous outcome definitions, all of which contributed to risk of bias and may inflate or attenuate individual effect estimates. Within these constraints, our synthesis supports several provisional practice considerations: (i) use short near-infrared wavelengths in the 700–850 nm range to ensure adequate tissue penetration; (ii) deliver treatment using a limited series of applications rather than a single exposure (approximately 4–10 sessions focused in the early postoperative period); (iii) favour non-contact application to avoid mechanical disruption of fresh wounds; and (iv) adapt dosing parameters to the anatomical site, recognising that superficially accessible, well-vascularized regions may respond differently than deeper or more complex surgical fields (Figure 3). Future trials should therefore standardise and clearly report anatomical site, control condition and key dosing parameters while minimising bias, so that these preliminary recommendations can be refined into formal clinical guidelines.

4.4 | Limitations

While this review provides evidence supporting the efficacy of NIR therapy for postoperative wound healing, several limitations should be acknowledged. First, the overall risk of bias across included trials was rated as 'some concerns', with recurring issues in randomization procedures, allocation concealment and blinding. These limitations are important because unclear allocation concealment can lead to systematic baseline differences, and incomplete blinding can inflate effects through performance and detection bias. This is particularly relevant for subjective endpoints such as pain scores and clinician-rated healing indices,

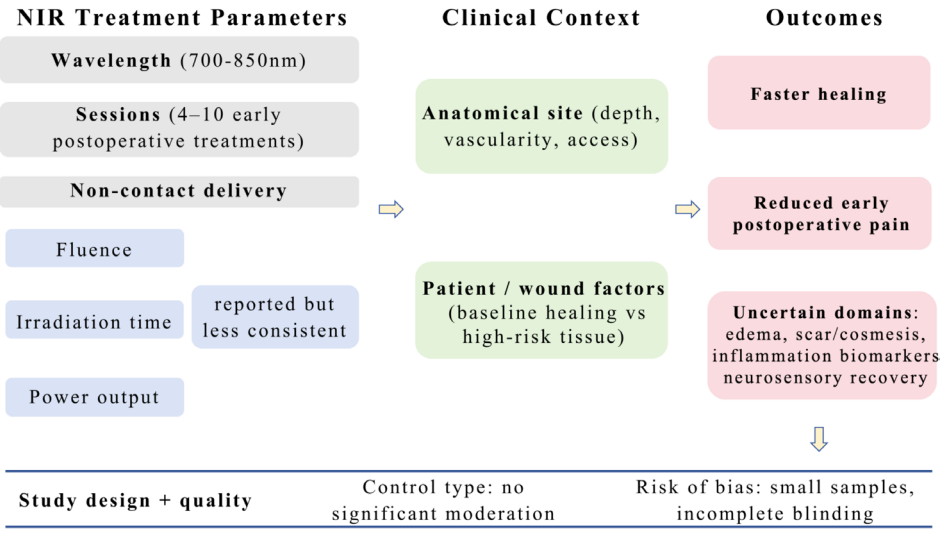


FIGURE 3 | Conceptual framework for optimising postoperative NIR therapy. The wavelength, sessions and non-contact delivery of NIR therapy are effective moderators for effective healing and reduced pain in surgical induced wounds. Fluence, irradiation time and power output of NIR therapy are not effective moderators.

which are more vulnerable to expectation effects than objective outcomes. These methodological limitations may introduce residual confounding into pooled estimates and may over—or underestimate true effects. Second, high between-study heterogeneity was observed in most analyses. This likely reflects variations in intervention parameters (e.g., wavelength range, delivered fluence, power output, number of sessions and application technique), as well as differences in outcome definitions, assessment timing and patient/surgical populations. Third, some eligible studies could not be quantitatively synthesised because of incomplete reporting (e.g., missing SD/SE data or incompatible formats), which limits the generalizability and may bias the pooled effects toward better-reported outcomes. Finally, several subgroup results—particularly for sensory/neurological recovery, cosmetic/scar outcomes and inflammatory biomarkers—were based on very few small studies and produced wide, imprecise confidence intervals; these signals should therefore be interpreted cautiously and not as definitive evidence of absence of benefit.

4.5 | Future Research

First, future trials should clearly report core NIR parameters—including wavelength, fluence, power output, irradiation time, number and timing of sessions, application technique (contact vs. non-contact), anatomical site and control condition—to ensure reproducibility and enable meaningful moderator analyses. More importantly, the field would benefit from adequately powered, multicentre, methodologically rigorous RCTs with standardised protocols, robust randomization and allocation concealment and, where feasible, sham controls and blinded outcome assessment, to reduce bias and provide more definitive estimates of clinical benefit. In addition, future trials should adopt more rigorous randomised designs, including prospective registration, prespecified primary outcomes, appropriate sample-size calculations, robust allocation concealment and sham-controlled blinding when feasible, to reduce bias and improve internal validity.

Second, work is needed to define optimal dosing rather than assuming that any NIR exposure is effective. Prior studies suggest that penetration depth (short NIR vs. superficial red light), single- versus dual-wavelength delivery, total fluence, number of sessions and application technique may all influence outcomes. Future studies should systematically vary these parameters in a controlled way to determine clinically effective wavelength-dose-frequency combinations. To improve generalizability and clinical relevance, these dose-optimization studies should include higher-risk populations in whom delayed healing is more common, such as older adults, patients with diabetes or metabolic syndrome, smokers and individuals with vascular compromise, and should report baseline risk factors and stratified or subgroup analyses where possible.

Third, most existing trials focus on short-term endpoints (early epithelialization, pain, edema). Future research should include longer-term and functionally meaningful outcomes such as scar quality, donor-site morbidity, neurosensory recovery and graft or flap stability, especially in under-studied anatomical regions. Longer follow-up should be paired with the inclusion of diverse surgical procedures and patient groups so that procedure-specific and population-specific effectiveness can be estimated more reliably.

5 | Conclusion

This systematic review and meta-analysis of 56 randomised controlled trials found that pooled estimates favoured NIR therapy for early postoperative outcomes, most notably (i) faster wound healing/tissue repair (0.78, [0.46–1.09], $p < 0.01$) and (ii) reduction of early postoperative pain (0.71, [0.24–1.17], $p < 0.01$). Because the evidence base was dominated by postoperative wounds in the oral cavity, these conclusions apply most directly to mucosal (predominantly oral/maxillofacial) wounds. However, effects were variable across trials, and the certainty of evidence was very low. Moderator analyses suggested larger

pooled effects when NIR was started intraoperatively or within 24–48 h after surgery and delivered with short NIR wavelengths (700–850 nm), non-contact application and moderate treatment schedules (about 4–10 sessions over the first 1–2 postoperative weeks), but these findings should be interpreted as hypothesis-generating. For cutaneous/skin wounds (e.g., external incisions and donor sites), the available evidence is limited and imprecise, and conclusions remain provisional because non-oral wound types were represented by relatively few trials. Overall certainty was rated very low because most trials had some concerns or high risk of bias, between-study heterogeneity was substantial ($I^2 \approx 89\%$), reporting was often incomplete, and publication bias could not be ruled out. Because heterogeneity remained high even within subgroups, the magnitude—and in some settings the presence—of benefit is uncertain, and the prediction interval in the primary analysis indicates that true effects may vary widely across clinical contexts, potentially ranging from little or no benefit to substantial benefit, and may even be negative in some settings. Effects on other domains—swelling/edema, scar/cosmetic outcomes, inflammatory biomarkers and neurosensory recovery—were small, imprecise, or based on few studies.

Taken together, these findings suggest that NIR therapy may offer benefit for early healing indices and postoperative pain in some settings, but current evidence is insufficient to recommend routine clinical use, and NIR should be considered investigational pending confirmation in rigorously designed, adequately powered, preregistered RCTs with standardised dosimetry, procedure-specific populations (including higher-risk patients), longer follow-up for scar outcomes and patient-centred endpoints.

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Ethics Statement

This is a systematic review and meta-analysis, and no participants were recruited in this study.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Characteristics of primary

studies. **Table S2:** List of excluded studies. **Figure S1:** Forest plot for postoperative wound healing. **Figure S2a:** Funnel Plot for postoperative pain. **Figure S2b:** Funnel Plot for postoperative wound healing. **Figure S3:** Forest plot of subgroup analysis of the wavelength effect. **Figure S4:** Forest plot of subgroup analysis of the fluence effect. **Figure S5:** Forest plot of subgroup analysis of the NIR power effect. **Figure S6:** Forest plot of subgroup analysis of the point time effect. **Figure S7:** Forest plot of subgroup analysis of the session time effect. **Figure S8:** Forest plot of subgroup analysis of the number of sessions effect. **Figure S9:** Forest plot of subgroup analysis of the intervention technique effect. **Figure S10:** Forest plot of subgroup analyses for clinical outcome domains after NIR therapy—pain and discomfort, wound healing and tissue repair, swelling, scar formation and cosmetic outcomes, immune response, sensory and neurological recovery. **Figure S11:** Forest plot of subgroup analyses by control condition for NIR therapy—comparison with standard postoperative care, placebo/sham light and microcurrent therapy. **Figure S12:** Forest plot of subgroup analyses by wound location for NIR therapy—perineum and genital area, oral cavity, eyelid, abdomen, inguinal, periauricular skin, chest, shoulder, neck, lip and donor sites for skin graft. **Figure S13:** Leave-one-out sensitivity analysis for the pooled effect of NIR therapy on surgical wound healing.